

Numerical methods

Lecture 6: On polynomials: estimates of the roots, Horner's method

Levente Lócsi

ELTE IK

- ① Estimates for the roots of polynomials
- ② Horner's method

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Estimates for the roots of polynomials

Let us consider polynomials of degree n having the form:

$$P(x) = a_n \cdot x^n + a_{n-1} \cdot x^{n-1} + \cdots + a_1 \cdot x + a_0$$
$$a_i \in \mathbb{R}, \quad a_0 \neq 0, \quad a_n \neq 0.$$

Remarks:

- We may allow $a_i \in \mathbb{C}$ too.
- If $a_0 = 0$, then $x = 0$ is a root, we can divide with $x \rightsquigarrow$ we can examine a simpler polynomial.
- If $a_n = 0$, then it isn't actually of degree n .

Estimates for the roots of polynomials

Example

Observe the location of the roots of some polynomials.
(Beware: complex roots may arise!)

Estimates for the roots of polynomials

Theorem: Estimates for the roots of polynomials

Given the polynomial $P(x) = a_n \cdot x^n + a_{n-1} \cdot x^{n-1} + \cdots + a_1 \cdot x + a_0$ with $a_0 \neq 0$ and $a_n \neq 0$, for each root x_k of P we have:

$$r < |x_k| < R,$$

with

$$R = 1 + \frac{\max_{i=0}^{n-1} |a_i|}{|a_n|}, \quad r = \left(1 + \frac{\max_{i=1}^n |a_i|}{|a_0|} \right)^{-1}.$$

Remark: Thus we have given an open ring on the complex plane centered at the origin which contains all the roots.

Estimates for the roots of polynomials

Proof.

- ① First we show that if $|x| \geq R$ (x is outside the outer circle), then $|P(x)| > 0$ (x is not a root of P). We may assume $|x| > 1$.

Using the triangle inequality:

$$a_n \cdot x^n = P(x) - \sum_{k=0}^{n-1} a_k \cdot x^k \implies |a_n \cdot x^n| \leq |P(x)| + \left| \sum_{k=0}^{n-1} a_k \cdot x^k \right|.$$

$$|P(x)| \geq |a_n x^n| - |a_{n-1} x^{n-1} + \dots + a_1 x + a_0|$$

To continue the estimation, increase to subtracted term:

$$\begin{aligned} |a_{n-1} x^{n-1} + \dots + a_0| &\leq |a_{n-1}| \cdot |x|^{n-1} + \dots + |a_0| \leq \\ &\leq \left(\max_{i=0}^{n-1} |a_i| \right) \cdot (|x|^{n-1} + \dots + 1) = \left(\max_{i=0}^{n-1} |a_i| \right) \cdot \frac{|x|^n - 1}{|x| - 1} < \\ &< \left(\max_{i=0}^{n-1} |a_i| \right) \cdot \frac{|x|^n}{|x| - 1}. \end{aligned}$$

Estimates for the roots of polynomials

(Proof continued.)

Continue the estimation of $|P(x)|$, and examine when is it positive.

$$|P(x)| > |a_n| \cdot |x|^n - \left(\max_{i=0}^{n-1} |a_i| \right) \cdot \frac{|x|^n}{|x| - 1} \geq 0$$

Reorder the inequality (multiply with $|x| - 1 > 0$ and divide with $|a_n| \cdot |x|^n$):

$$\begin{aligned} |P(x)| > 0 &\iff |a_n| \cdot |x|^n \geq \left(\max_{i=0}^{n-1} |a_i| \right) \cdot \frac{|x|^n}{|x| - 1} \\ \iff |x| - 1 &\geq \left(\max_{i=0}^{n-1} |a_i| \right) \cdot \frac{|x|^n}{|a_n| \cdot |x|^n} \\ \iff |x| &\geq 1 + \frac{\max_{i=0}^{n-1} |a_i|}{|a_n|} =: R. \end{aligned}$$

Thus if $|x| > R$, then $|P(x)| > 0$, i.e. x is not a root.

Estimates for the roots of polynomials

(Proof continued.)

- ② For the lower estimate, we will use the above proved upper estimate for the reciprocal polynomial of P .

Introduce $y := \frac{1}{x}$ ($x \neq 0$):

$$\begin{aligned} P(x) &= P\left(\frac{1}{y}\right) = a_n \left(\frac{1}{y}\right)^n + a_{n-1} \left(\frac{1}{y}\right)^{n-1} + \dots + a_1 \left(\frac{1}{y}\right) + a_0 = \\ &= \left(\frac{1}{y}\right)^n \cdot \underbrace{(a_n + a_{n-1}y + \dots + a_1 y^{n-1} + a_0 y^n)}_{Q(y)} = x^n \cdot Q\left(\frac{1}{x}\right). \end{aligned}$$

The polynomial Q is called the *reciprocal polynomial* of P .

$$P(x_k) = 0 \quad \Leftrightarrow \quad Q\left(\frac{1}{x_k}\right) = 0,$$

I.e. the roots of Q are the reciprocals of the roots of P .

Estimates for the roots of polynomials

(Proof continued.)

Apply the previous upper estimate for Q :

$$\frac{1}{|x_k|} < 1 + \frac{\max_{i=1}^n |a_i|}{|a_0|} =: \frac{1}{r} \quad \Longleftrightarrow \quad |x_k| > r.$$



Remark: We can allow polynomials with complex coefficients in the theorem too, it would not alter the method of the proof. However we may think of polynomials with just real coefficients, but complex roots can arise anyway.

① Estimates for the roots of polynomials

② Horner's method

for the calculation of function values and derivative values of polynomials at given points.

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

Organize the terms, pull out x where possible:

$$\begin{aligned} P(x) &= \underbrace{(a_n x^{n-1} + a_{n-1} x^{n-2} + \dots + a_2 x + a_1)}_{a_1^{(0)}} \cdot x + a_0 = \\ &= \underbrace{((a_n x^{n-2} + a_{n-1} x^{n-3} + \dots + a_2)) \cdot x + a_1}_{a_2^{(0)}} \cdot x + a_0 = \\ &= \dots = \underbrace{(\dots (a_n x + a_{n-1})) \cdot x + \dots}_{a_{n-1}^{(0)}} \cdot x + a_0. \end{aligned}$$

Remark. Also known as: Horner algorithm, Horner scheme.

Definition: Horner's method

The function value at the point ξ of the polynomial $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ can be calculated using the following algorithm:

① $a_n^{(0)} := a_n,$

② $a_k^{(0)} := a_k + \xi \cdot a_{k+1}^{(0)} \quad (k = n-1, \dots, 1, 0)$

and $P(\xi) = a_0^{(0)}.$

Proposition: Operation count for Horner's method

A polynomial of degree n can be evaluated at a given point using n multiplications and n additions, i.e. in $\mathcal{O}(n)$ time.

Proof. ✓



Table to evaluate $P(\xi)$ by hand:

a_n	a_{n-1}	a_{n-2}	$\dots$	a_k	$\dots$	a_1	a_0
ξ	$\xi \cdot a_n^{(0)}$	$\xi \cdot a_{n-1}^{(0)}$	$\dots$	$\xi \cdot a_{k+1}^{(0)}$	$\dots$	$\xi \cdot a_2^{(0)}$	$\xi \cdot a_1^{(0)}$
$a_n^{(0)}$	$a_{n-1}^{(0)}$	$a_{n-2}^{(0)}$	$\dots$	$a_k^{(0)}$	$\dots$	$a_1^{(0)}$	$a_0^{(0)}$

Example

Evaluate the polynomial $P(x) = x^5 + 6x^4 - x^3 + 3x^2 - 15x - 7$ at $\xi = 2$ using Horner's method!

$$P(x) = x^5 + 6x^4 - x^3 + 3x^2 - 15x - 7, \quad \xi = 2$$

1	6	-1	3	-15	-7
2	2 · 1	2 · 8	2 · 15	2 · 33	2 · 51
1	8	15	33	51	95

$$\begin{aligned}
 P(2) &= 2^5 + 6 \cdot 2^4 - 2^3 + 3 \cdot 2^2 - 15 \cdot 2 - 7 = \\
 &= 32 + 96 - 8 + 12 - 30 - 7 = 95
 \end{aligned}$$

Proposition: Horner's method and the derivative

The polynomial P can be written in the following form:

$$P(x) = a_0^{(0)} + (x - \xi) \cdot \overbrace{(a_n^{(0)} x^{n-1} + \cdots + a_1^{(0)})}^{P_1(x)}$$

with coefficients $a_i^{(0)}$ ($i = 0, \dots, n$) given by Horner's algorithm.
Furthermore

$$P'(\xi) = P_1(\xi).$$

Remark. $\sim$ Taylor polynomial centered at ξ .

Horner's method and the derivative

$$P(x) = a_0^{(0)} + (x - \xi) \cdot \underbrace{(a_n^{(0)} x^{n-1} + \dots + a_{k+1}^{(0)} x^k + a_k^{(0)} x^{k-1} + \dots + a_1^{(0)})}_{P_1(x)}$$

Proof.

- 1 the coefficient of x^k ($k = 0, \dots, n-1$) in P
(x^n separately: HW)
 - on one hand, by definition: a_k ,
 - on the other, hand as above: $a_k^{(0)} - \xi \cdot a_{k+1}^{(0)}$.
 - by Horner's algorithm: $a_k^{(0)} = a_k + \xi \cdot a_{k+1}^{(0)}$. ✓
- 2 The derivative of P from the above form (sum, product rules):

$$P'(x) = 1 \cdot P_1(x) + (x - \xi) \cdot P_1'(x) \implies P'(\xi) = P_1(\xi).$$



Horner's method and the derivative

We use Horner's algorithm again to compute $P_1(\xi)$ with coefficients $a_n^{(0)}, \dots, a_1^{(0)}$.

$$\textcircled{1} \ a_n^{(1)} := a_n^{(0)},$$

$$\textcircled{2} \ a_k^{(1)} := a_k^{(0)} + \xi \cdot a_{k+1}^{(1)} \quad (k = n-1, \dots, 1),$$

and $P_1(\xi) = P'(\xi) = a_1^{(1)}$.

Continue the table:

a_n	a_{n-1}	a_{n-2}	$\dots$	a_1	a_0
$\boxed{\xi}$	$\xi \cdot a_n^{(0)}$	$\xi \cdot a_{n-1}^{(0)}$	$\dots$	$\xi \cdot a_2^{(0)}$	$\xi \cdot a_1^{(0)}$
$a_n^{(0)}$	$a_{n-1}^{(0)}$	$a_{n-2}^{(0)}$	$\dots$	$a_1^{(0)}$	$\boxed{a_0^{(0)} = P(\xi)}$
$\boxed{\xi}$	$\xi \cdot a_n^{(1)}$	$\xi \cdot a_{n-1}^{(1)}$	$\dots$	$\xi \cdot a_2^{(1)}$	
$a_n^{(1)}$	$a_{n-1}^{(1)}$	$a_{n-2}^{(1)}$	$\dots$	$\boxed{a_1^{(1)} = P_1(\xi)}$	

Horner's method and Taylor

We can continue even further...

$$P(x) = a_0^{(0)} + (x - \xi) \cdot P_1(x)$$

Proposition: Horner's method and higher derivatives

The polynomial P can be written in the following form:

$$P(x) = a_0^{(0)} + a_1^{(1)}(x - \xi) + a_2^{(2)}(x - \xi)^2 + \cdots + a_n^{(n)}(x - \xi)^n$$

with coefficients $a_i^{(j)}$ ($j = 0, \dots, n$; $i = j, \dots, n$) given by Horner's algorithm. Furthermore

$$\frac{P^{(j)}(\xi)}{j!} = P_j(\xi) = a_j^{(j)}$$

with $P_j(x) = a_n^{(j-1)}x^{n-j} + \cdots + a_j^{(j-1)}$.

Proof. By induction, skipped.

Remark. Continuing the table until we have only one element, we will find the coefficients of the Taylor polynomial of P centered at ξ in the diagonal.

Example

Find the Taylor polynomial centered at $\xi = 1$ of the polynomial $P(x) = x^3 - x^2 + x - 1$ using Horner's method!

$$P(x) = x^3 - x^2 + x - 1, \quad \xi = 1$$

$$\begin{array}{r}
 1 \quad -1 \quad 1 \quad -1 \\
 \boxed{1} \quad 1 \quad 0 \quad 1 \\
 \hline
 1 \quad 0 \quad 1 \quad \boxed{0} \\
 \boxed{1} \quad 1 \quad 1 \\
 \hline
 1 \quad 1 \quad \boxed{2} \\
 \boxed{1} \quad 1 \\
 \hline
 1 \quad \boxed{2} \\
 \boxed{1} \\
 \hline
 \boxed{1}
 \end{array}$$

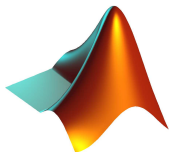
$$P(1) = 0$$

$$P'(x) = 3x^2 - 2x + 1, \quad P'(1) = 2$$

$$P''(x) = 6x - 2, \quad P''(1) = 4$$

$$P'''(x) = 6, \quad P'''(1) = 6$$

$$(T_{3,1}P)(x) = 0 + 2 \cdot (x-1) + 2 \cdot (x-1)^2 + 1 \cdot (x-1)^3$$



- 1 Roots and estimates thereof in case of some polynomials with random (real or complex) coefficients of higher degrees ($n = 5, 10, 50, 100$).